Measurement of π^0 and η Meson Production in $p + p$ Collisions at $\sqrt{s} = 200$ GeV With sPHENIX

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This dissertation titled

Measurement of π^0 and η Meson Production in $p + p$ Collisions at $\sqrt{s} = 200$ GeV With sPHENIX

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Abstract

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Measurement of π^0 and η Meson Production in $p + p$ Collisions at $\sqrt{s}$ = 200 GeV With sPHENIX (32 pp.)

Director of Dissertation: Justin E. Frantz

This dissertation presents a study of neutral meson production in proton-proton ($p + p$) collisions at $\sqrt{s}$ = 200 **G**iga **E**lectron **V**olt (GeV) collected by newly built **s**uper **P**ioneering **H**igh **E**nergy **N**uclear **I**nteraction e**X**periment (sPHENIX) detector. sPHENIX detector, within **R**elativistic **H**eavy **I**on **C**ollider (RHIC) facility, is a state-of-the-art successor to **P**ioneering **H**igh **E**nergy **N**uclear **I**nteraction e**X**periment (PHENIX) with extended pseudorapidity (η) and complete ϕ-coverage. Our research focuses on the measurement of invariant cross section of η meson and π^0 via their two-photon decay mode. The final results are compared with results from PHENIX experiment to provide the initial validation of data measured by sPHENIX detector. In particular, a first full physics measurement of the ratio of η meson to π^0 production cross sections is made and found to be in good agreement with past PHENIX measurement.

Furthermore, this dissertation reports various background reconstruction strategies and the application of supervised machine learning to isolate η meson signals from background. Ultimately, we discuss the implementation of calibration methods of sPHENIX **E**lectro**M**agnetic **Cal**orimeter (EMCal) using π^0 decaying to two photons, as well as the methods' development and validation.

Dedication

Dedicated to the perseverance that outlasted every setback, the discipline that prevailed over every distraction, and the countless sleepless nights that illuminated this journey marked by curiosity, resilience, and growth.

Acknowledgements

Several AI tools proved to be valuable, especially during the last few years of this journey. The most significant were Notebook LM (for reading papers), Google Colaboratory (Python computing), Google Gemini (for image and video generation), Perplexity AI (as a search engine), and last but not least "ChatGPT", which was the master of all.

Finally, I am forever grateful to the city of Athens, which has been my "home-away-from-home" for the past seven years. I would also like to remember NEPSA, ISU, ISSS (especially to ISSS director Diane Cahill), SAC, and HAPCAP for providing me with volunteering opportunities in different positions. Such experiences played a significant role in my personal growth. Upon leaving Athens, I realize that I am profoundly different from the person who first arrived here in 2019. And yeah, coming from Nepal, the "mountains" around Athens are basically adorable little speed bumps ☺.

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List of Symbols

α	energy asymmetry
α_s	strong coupling constant
χ^2	shower shape variable
ΔR	angular separation in η–ϕ space
ϵ	detector efficiency
η	pseudorapidity
η meson	eta meson
γ	photon
Λ_{QCD}	QCD scale parameter
ϕ	azimuthal angle
π^0	neutral pion
$\sqrt{s}$	proton-proton center-of-mass energy
$\sqrt{s_{\mathrm{NN}}}$	nucleon-nucleon center-of-mass energy
θ	polar angle
$\theta_{\gamma\gamma}$	opening angle
A	detector acceptance
$m_{\gamma\gamma}$	di-photon invariant mass
$mrad$	milliradian
p_T	transverse momentum
Q	momentum transfer
z_{vtx}	z - component of collision vertex

List of Acronyms

AGS **A**lternating **G**radient **S**ynchrotron
EMCal **E**lectro**M**agnetic **Cal**orimeter
GeV **G**iga **E**lectron **V**olt
INTT **INT**ermediate Silicon **T**racker
MeV **M**ega **e**lectron **V**olt
MVTX **M**onolithic Active Pixel Sensors **V**er**T**e **X**Detector
PHENIX **P**ioneering **H**igh **E**nergy **N**uclear **I**nteraction e**X**periment
pp **p**roton - **p**roton
QCD **Q**uantum **C**hromo**D**ynamics
RHIC **R**elativistic **H**eavy **I**on **C**ollider
sPHENIX **s**uper **P**ioneering **H**igh **E**nergy **N**uclear **I**nteraction e**X**periment
TPC **T**ime **P**rojection **C**hamber

1 Introduction and Physics Motivation

This dissertation begins with the coordinate system and kinematic variables used in collider experiments. They are followed by an overview of the relativistic heavy-ion collisions, and the sPHENIX detector. The chapter concludes with the motivation and scope of this dissertation.

1.1 Coordinate System and Kinematic Variables

In this section, we introduce some of the technical terms and concepts commonly used in high-energy nuclear physics.

1.1.1 Pseudorapidity (η)

Pseudorapidity (η) is a spatial coordinate commonly used in collider physics to describe the angle of a particle relative to the beam axis. It is defined as:

$$\eta = -\ln\left[\tan\left(\frac{\theta}{2}\right)\right] \tag{1.1}$$

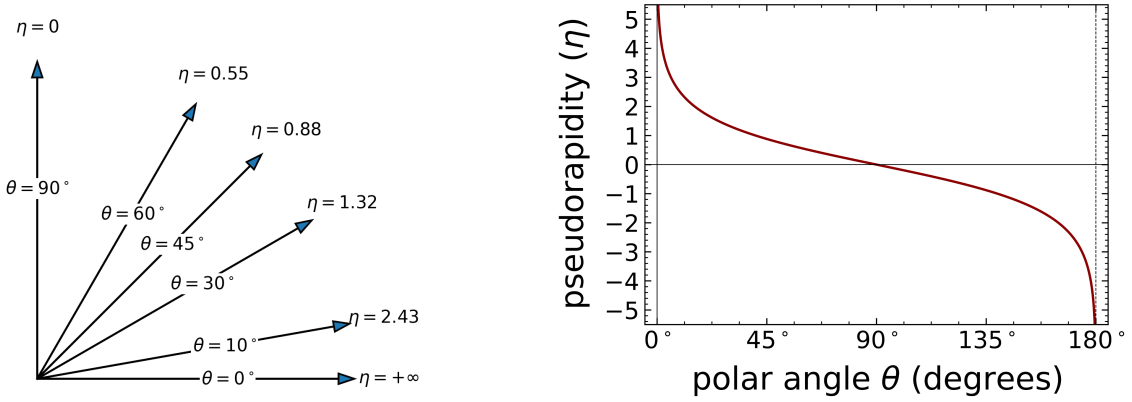


Figure 1.1: Relationship between polar angle (θ) and pseudorapidity (η): (a) geometric interpretation showing η values for different polar angle θ ; (b) functional dependence of η on θ, illustrating its divergence as $\theta \to 0°$ and $\theta \to 180°$. Both images were created using Python.

From equation 1.1, it is evident that pseudorapidity (η) is different mathematical form of the polar angle (θ). Still, pseudorapidity is generally preferred over the polar angle as pseudorapidity (η) is closer to being a Lorentz invariant quantity. Pseudorapidity (η) acts as a proxy for rapidity (y) and achieves longitudinal boost invariance in the high-energy, massless limit. The relationship between pseudorapidity (η) and polar angle (θ) is also reflected in the figure 1.1.

2 Theoretical Framework

2.1 Particle Physics and the Standard Model

Particle physics studies the elementary building blocks of matter, radiation and the interactions between them. So far, there are six quarks, six leptons, four gauge bosons types, and one Higgs boson within the family of elementary particles, whose dynamics are governed by symmetries and conservation laws. Elementary particles help to build the rich structure of nuclei, atoms, and macroscopic matter. Similarly, the forces between particles arise from the exchange of gauge bosons and are associated with the local symmetries [Gri08].

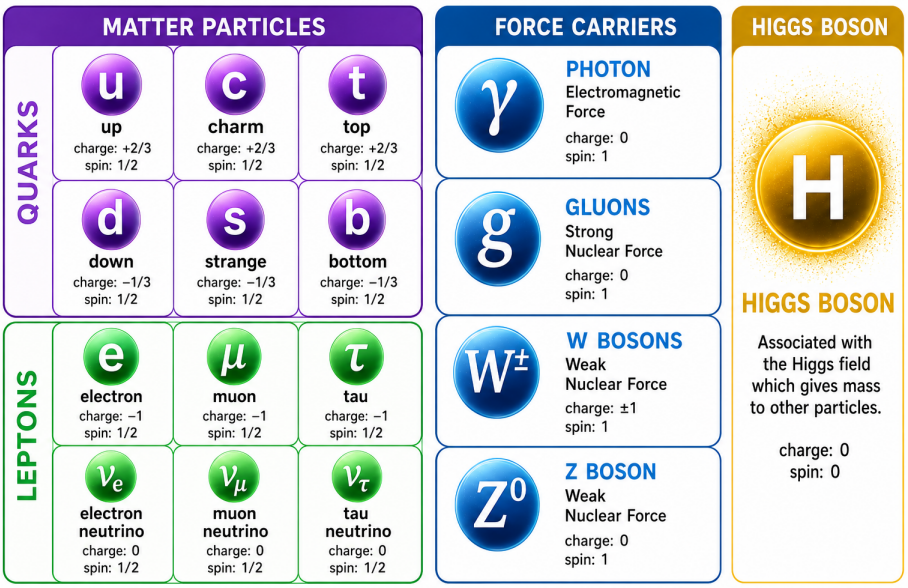


Figure 2.1: A schematic overview of the Standard Model showing fundamental particles and force carriers. This image was generated with the help of ChatGPT.

The "Standard Model" organizes the elementary particles into three categories: matter **fermions**, **gauge bosons**, and the **Higgs boson**. As a reference from figure 2.1, quarks and leptons makes fermions, while all the force carrier makes gauge bosons. There are

six different flavors (like u, d, s, c, b, t) of quarks and all of them carry color charge, and participate in strong, weak and electromagnetic interaction. The leptons (e, μ, τ) and their corresponding neutrinos are colorless and do not interact via strong force [HM84, Gri08].

2.2 Quantum Chromodynamics (QCD)

Quantum Chromodynamics (QCD) is the theory of the strong interaction in the Standard Model. It describes the behavior of quarks and gluons through the exchange of color charge [PS95, Gri08]. **Q**uantum **C**hromo**D**ynamics (QCD) is a non-Abelian gauge theory built on the $SU(3)_C$ color symmetry group, where gluons not only mediate force between quarks but also interact among themselves [PS95].

2.2.1 QCD Lagrangian

The dynamics of quarks and gluons are encoded in the QCD Lagrangian density, which for N_f quark flavors can be written as [LB14, ESW96, PS95]

$$\mathcal{L}_{\text{QCD}} = \sum_{f=1}^{N_f} \bar{\psi}_f \left(i\gamma^\mu D_\mu - m_f \right) \psi_f - \frac{1}{4} F^a_{\mu\nu} F^{\mu\nu}_a \tag{2.1}$$

where ψ_f denotes the quark field of flavor f, m_f is the corresponding quark mass, and D_μ is the covariant derivative that couples quarks to the gluon gauge fields. The gluon field strength tensor $F^a_{\mu\nu}$ describes both the propagation of gluon fields and their self-interactions, reflecting the non-Abelian nature of the $SU(3)_c$ gauge symmetry [LB14, ESW96, PS95].

2.2.2 The Running Coupling Constant

The strong coupling constant (α_s) is the fundamental parameter governing the strength of the strong interaction in Quantum Chromodynamics. Analogous to the electromagnetic fine structure constant "$\alpha_{\text{em}} = e^2/(4\pi)$", it is defined in terms of the QCD gauge coupling "g_s" as [DBdT16]

$$\alpha_s(\mu^2) \equiv \frac{g_s^2(\mu^2)}{4\pi}, \tag{2.2}$$

where μ denotes the renormalization scale.

The equation 2.2, being an example of non Abelian theory, have the scale dependence feature. This means "α_s" in equation 2.2 cannot be determined exactly and must be approximated separately in different energy scale. When $Q^2 > \Lambda^2_{\text{QCD}}$, we use perturbation expansion and the equation 2.2 takes the following form [DBdT16]:

$$\alpha_s(\mu^2) = \frac{4\pi}{\beta_0 \ln\left(\mu^2/\Lambda^2_{\text{QCD}}\right)}, \tag{2.3}$$

where Λ_{QCD} is the QCD scale parameter. In applications to physical processes characterized by a momentum scale Q, it is common to choose the renormalization scale $\mu^2 = Q^2$, so that the coupling is written as $\alpha_s(Q^2)$. In the regime where $Q^2 \gg \Lambda^2_{\text{QCD}}$, the logarithm in the denominator of the equation 2.3 becomes large enough to make the coupling constant parametrically small, ($\alpha_s(Q^2) \ll 1$). Therefore, at much higher Q^2 region (high energy or short distance), QCD interactions become weak and develops the property of asymptotic freedom.

3 Experimental Setup

The analysis presented in this dissertation was performed using data taken by sPHENIX detector. The detector was located within the RHIC complex at Brookhaven National Laboratory. It was installed at 8 o'clock interaction point of the RHIC ring, replacing its predecessor "PHENIX" detector [Os24, B$^+$24]. This chapter begins with an overview of the RHIC complex, particle acceleration (for both protons and heavy ions) proceeds with a summary of sPHENIX detector, and ends with a detailed explanation of EMCal subsystem.

3.1 The RHIC Complex

The RHIC accelerator was built on the legacy of the **A**lternating **G**radient **S**ynchrotron (AGS), later of which was operational since 1960s [HLO03, USD15]. The AGS revolutionized the concept of particle acceleration by using the principle of strong focusing (or Alternate Gradient). It utilized an alternating sequence of quadrupole electromagnets to constrain the transverse spread of the particle beam and achieve the unprecedented energy in particle acceleration [Ame11].

3.2 Proton and Heavy-Ion Acceleration at RHIC

Although both protons and heavy ions (such as Au) utilize same RHIC accelerator facility, their acceleration processes involve fundamentally different approaches. Heavy-ion acceleration focuses on progressive charge stripping, control of space-charge effects and management of intrabeam scattering and emittance growth [HLO03]. In contrast, proton acceleration focuses on preserving and manipulating the beam polarization for spin-physics experiments [HLO03]. The proton acceleration requires specialized polarized ion sources, spin rotators, and Siberian snake magnets (in AGS and RHIC rings) to overcome depolarizing resonances [HLO03].

4 Electromagnetic Calorimeter Calibration

The EMCal calibration is essential to ensure the measured energy by calorimeter towers reflects the true particle energy after considering detector effects. There are different calibration methods (like tower-slope method, absolute method, etc.) applied sequentially to EMCal. In this section, we discuss the absolute EMCal calibration and its implementation across different collision systems.

4.1 Overview of Calibration Strategy

Once cluster objects are created, we make all possible cluster pairs within every event. For each unique pair, we perform four vector addition to make an invariant mass distribution. We can observed peak approximately around 135 **M**ega **e**lectron **V**olt (MeV) corresponding to π^0 rest mass.

4.2 Implementing π^0 Calibration

sPHENIX took first data in 2023 and the π^0 calibration has been consistently implemented across different collision systems. During its operational lifetime, sPHENIX has collected many dataset across $Au+Au$, $O+O$ and $p+p$ collisions. At the time of writing this dissertation, we have successfully implemented initial π^0 calibration on three datasets.

As analysis presented in this dissertation was based on Run 24 **p**roton - **p**roton (pp) data, the calibration corresponding to this data set was used throughout the analysis. The final calibration value after six iteration of EMCal calibration was used in our analysis. The tower by tower calibration was performed and those tower information were passes to "clustering module" to make cluster object. Those cluster objects were later used for our physics analysis.

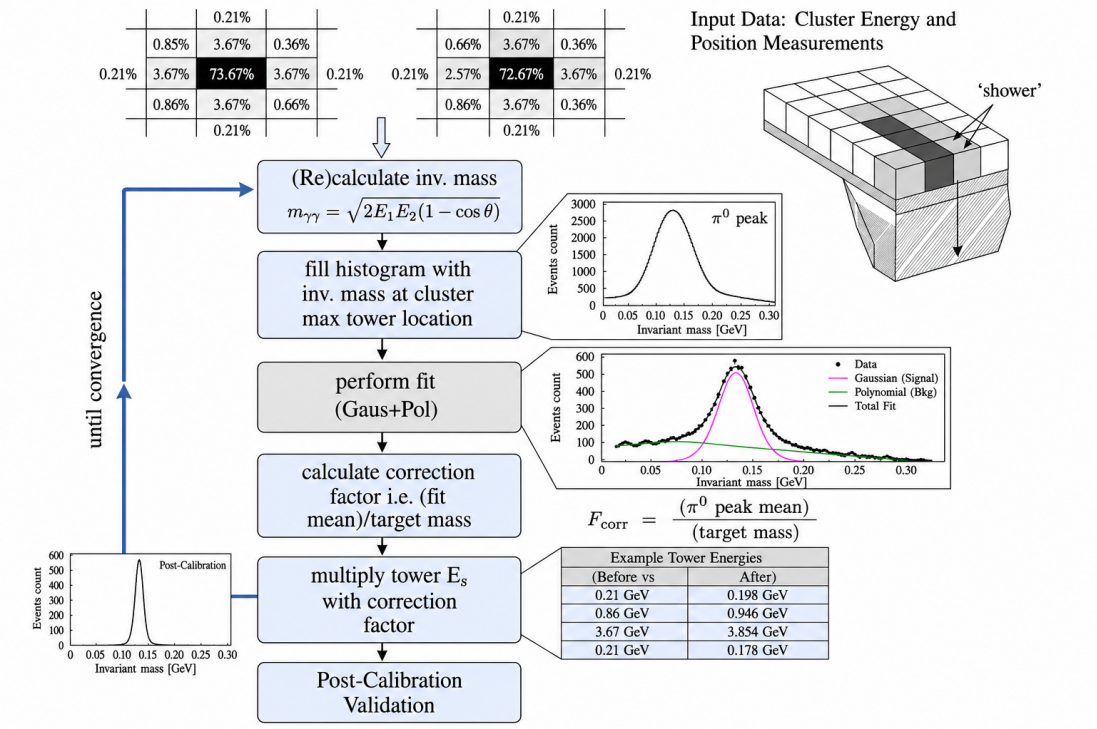


Figure 4.1: The schematic illustration of workflow for EMCal calibration based on two photons invariant mass reconstruction. Photon cluster pairs are used to reconstruct the invariant mass and π^0 peak is fitted to ultimately determine tower-dependent energy correction factors. This procedure is repeated until the reconstructed π^0 mass converges to the target value. Image credit: Figure generated with combined assistance from Google Gemini and ChatGPT.

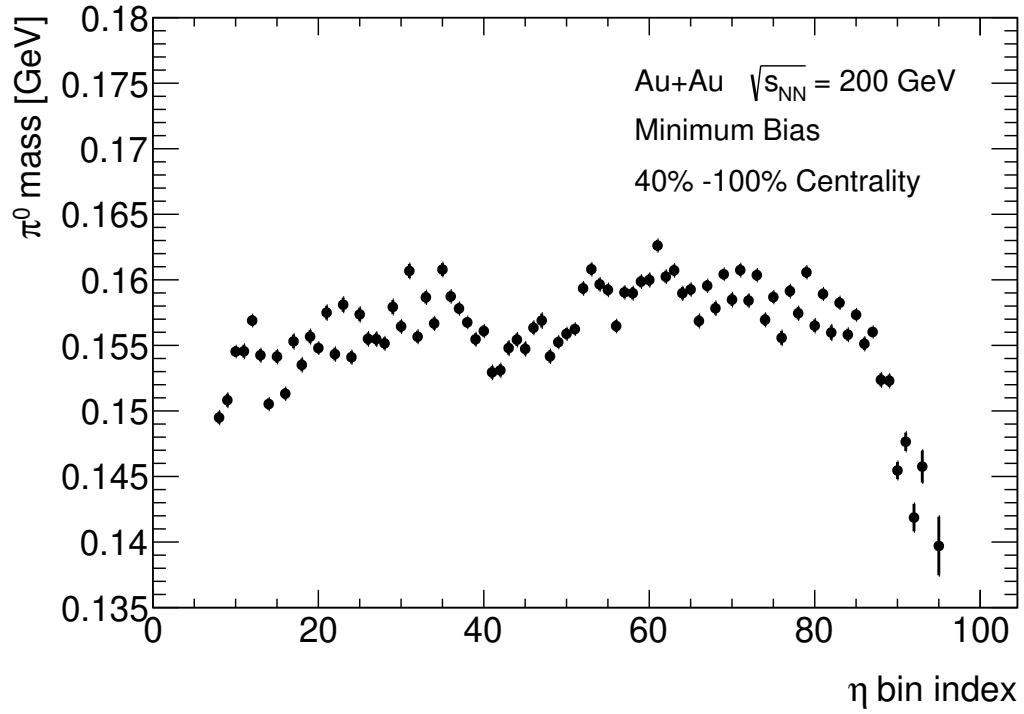


Figure 4.2: Reconstructed π^0 mass as a function of η bin index in Run 23 *Au* + *Au* data for first iteration while implementing absolute EMCal calibration process. One unit of η bin index corresponds to one tower along η direction.

5 Neutral Meson Analysis

5.1 Data Sets and Event Selection

This study utilized 698 different runs corresponding to zero *mrad* crossing angle from Run 24 proton-proton collisions. In the context of RHIC data taking, the term *Run* refers to a data taking campaign corresponding to a specific calendar year. Each store is divided into multiple (sometime single) time periods of continuous data taking intervals called *runs*. These intervals are determined by the stability of detector and data acquisition and never exceeds one hour time duration.

5.1.1 Run Selection

For the study of neutral meson both as a function of transverse momentum (p_T) and pseudorapidity (η), we use Run 24 $p + p$ collision from run 47289 to run 51237. They correspond to zero *mrad* crossing angle and have wide vertex distribution, thus, perfect candidate runs for our study. The analysis presented in this dissertation used the data produced with build tag *ana509_2024p022_v001*, which had following physics characteristics:

- Only physics runs were included. Pedestal and Cosmic runs were excluded
- Magnet status: ON
- Minimum run duration: > 300 s
- Detector QA status: GOLDEN
- Minimum Bias Trigger must be present
- Livetime threshold: $> 70\%$
- Minimum number of reconstructed events: $> 10^6$ (1 M) events

5.1.2 Event Selection

For our analysis involving transverse momentum (p_T), events which have reconstructed primary vertex position satisfying $|z_{\text{vtx}}| < 30$ cm were selected. Similarly, while performing analysis as a function of pseudorapidity (η), events with wider vertex ($|z_{\text{vtx}}| < 200$ cm) were considered. Another event level selection criteria involved trigger conditions. We used three different trigger data sets in this analysis: the minimum bias (MB) trigger, the Photon 3 GeV trigger, and the Photon 4 GeV trigger.

Although we used same number of runs, the integrated luminosity for pseudorapidity (η) based analysis is different than that for transverse momentum (p_T) based analysis. The integrated luminosities for the data sets (for all types of trigger conditions) used for transverse momentum p_T dependent analysis are summarized in Table 5.1.

Table 5.1: Event statistics and corresponding integrated luminosities for all trigger types that we use for p_T dependent analysis using Run 24 $p + p$ data.

Trigger Sample	Number of Events ($\times 10^9$)	Integrated Luminosity (pb^{-1})
Minimum Bias	0.5540	0.0244219
Photon 3 GeV	0.5110	5.856
Photon 4 GeV	0.5096	18.026

6 Results

6.1 Overview

In this chapter, we present the results of our analysis. This consists of the invariant cross section for π^0 and η mesons using Run 24 pp data, both as functions of p_T and pseudorapidity (η). The invariant cross section, along with the η to π^0 ratio, was compared with previous PHENIX measurements wherever applicable. As discussed in previous chapters, the results were obtained using a combination of minimum bias and two types of photon-triggered data sets.

6.2 Invariant Mass Peak and Width

The first measurement was the mass (signal peak) for both π^0 and η mesons. We measured peak positions as a function of transverse momentum (p_T) to verify the consistency of mass (peak mean) and sigma (width). To calculate this, we constructed an invariant mass histogram using golden parameter cuts. Then, using a side-band fitting process, estimate complete background and subtracted it to get a signal distribution. The final step we did was to fit the signal distribution using a Gaussian function to extract the mean and sigma of the meson peak. The range of fitting gaussian function was $0.06 \leq m_{\gamma\gamma} \leq 0.25$ for π^0 and $0.38 \leq m_{\gamma\gamma} \leq 0.76$ for η mesons.

The figure 6.1 suggest that the mass and sigma distributions are consistent as a function of p_T and across all three triggered data. In Figure 6.1, we can see sharp rise in mass and sigma for $p_T \geq 6$ GeV. This effect is due to merging behavior of π^0 photons, which limits our π^0 analysis at very high p_T.

Similarly, η mesons show a small and gradual drop in both mass and sigma on increasing p_T. This is caused by our method of signal extraction and a better S/B ratio in the higher p_T range. However, this difference is insignificant (just a few percent), so we can assume constant behavior of mass and sigma for the η meson as well.

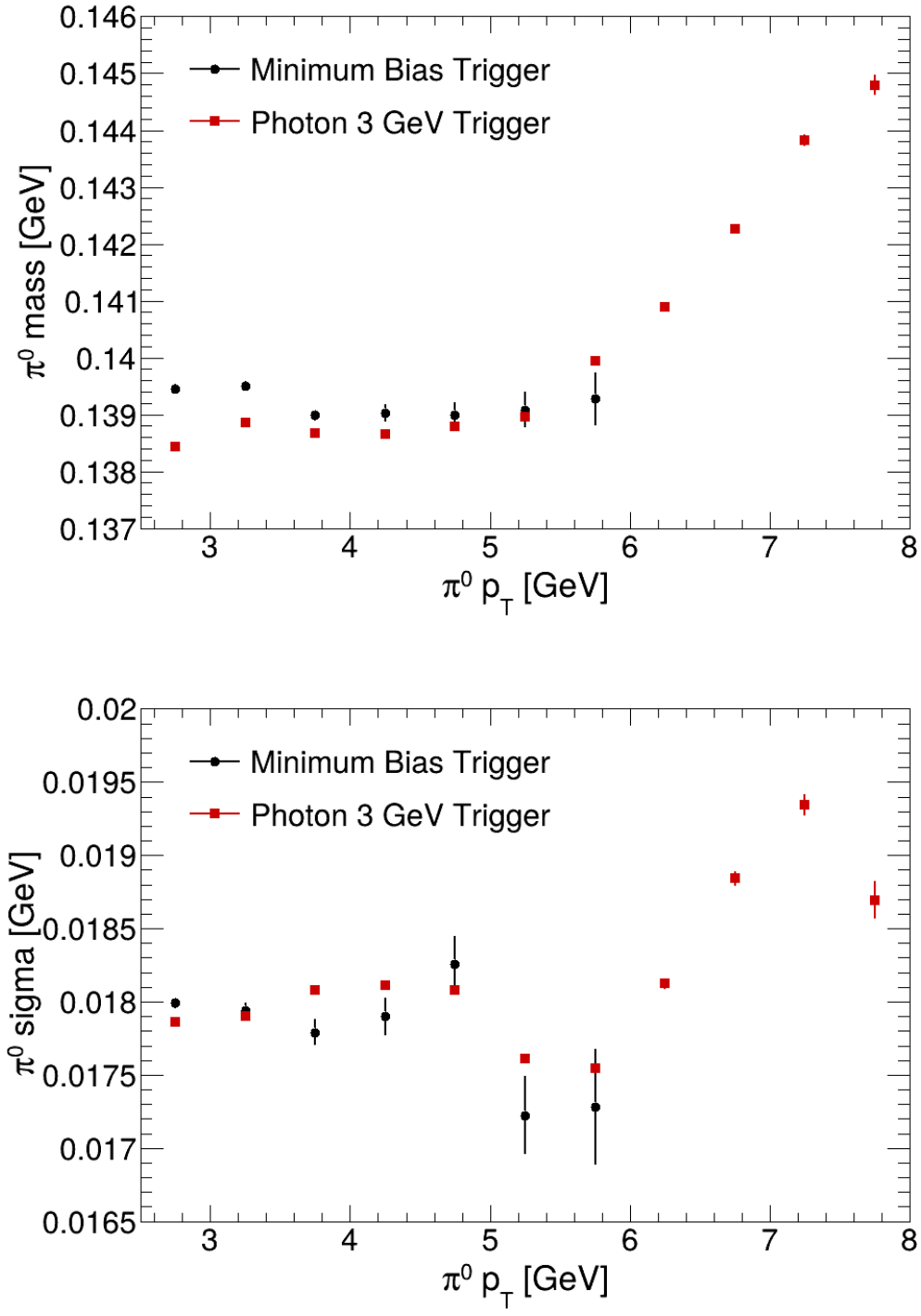


Figure 6.1: Distribution of π^0 mass (**top panel**) and sigma (**bottom panel**) as a function of transverse momentum (p_T). The error in both plots are obtained from the fitting procedure.

7 Discussion and Conclusion

7.1 Summary of Results

In this dissertation, a broad study of neutral meson production in proton-proton collisions at $\sqrt{s} = 200$ GeV was presented using sPHENIX data. The analysis focused on the measurement of π^0 and η mesons via two photon decay channels and reconstructed within the electromagnetic calorimeter. Furthermore, the implementation of absolute calibration of EMCal was also discussed. The Ohio University group was one of the primary group responsible for the development and implementation of EMCal absolute calibration within sPHENIX.

Invariant differential cross sections for π^0 and η mesons were measured as functions of transverse momentum (p_T) over a broad kinematic range. The results were obtained using a combination of minimum bias and photon-triggered data, enabling access to both low and high p_T regions. In addition, the η/π^0 ratio was evaluated as a function of transverse momentum, providing a complementary observable with reduced systematic uncertainties. Both the invariant cross section and η/π^0 ratio were evaluated for different acceptance regions, $|\eta| < 0.35$ and $|\eta| < 1.0$, both for comparison with PHENIX and with full sPHENIX pseudorapidity coverage.

Similarly, the analysis (η to π^0 ratio) was extended to study the pseudorapidity dependence of meson production using Photon 4 GeV triggered data. Measurements was performed in selected transverse momentum intervals, allowing for a detailed exploration of longitudinal aspects of hadron production.

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Appendix A: Run 24 $p + p$ Data Set: Selected Runs

The following run numbers corresponding to the Run 24 $p + p$ data set were used in this analysis. These runs were selected from golden run list available and corresponds to zero crossing angle. Following full run list is provided for reproducibility.

47289, 47291, 47293, 47306, 47310, 47313, 47314, 47315, 47316, 47333, 47334, 47357, 47360, 47363, 47376, 47377, 47378, 47379, 47380, 47381, 47382, 47391, 47393, 47394, 47395, 47396, 47440, 47443, 47454, 47455, 47457, 47458, 47459, 47461, 47462, 47463, 47464, 47474, 47478, 47480, 47481, 47482, 47484, 47485, 47489, 47491, 47492, 47494, 47495, 47497, 47501, 47502, 47503, 47505, 47506, 47507, 47509, 47513, 47514, 47516, 47517, 47519, 47521, 47522, 47524, 47525, 47540, 47547, 47548, 47549, 47552, 47557, 47568, 47634, 47635, 47636, 47637, 47638, 47650, 47657, 47658, 47659, 47661, 47662, 47664, 47666, 47667, 47714, 47715, 47716, 47719, 47720, 47721, 47722, 47723, 47724, 47725, 47726, 47727, 47729, 47730, 47732, 47733, 47783, 47831, 47834, 47889, 47890, 47892, 47893, 47940, 47944, 47946, 47966, 47972, 47973, 47975, 47976, 47977, 47978, 47979, 47982, 47998, 47999, 48001, 48002, 48008, 48073, 48076, 48079, 48082, 48084, 48085, 48086, 48088, 48089, 48097, 48099, 48166, 48172, 48176, 48177, 48179, 48181, 48185, 48230, 48233, 48638, 48639, 48640, 48645, 48656, 48657, 48658, 48660, 48701, 48706, 48720, 48721, 48722, 48725, 48726, 48727, 48730, 48731, 48732, 48734, 48735, 48736, 48742, 48743, 48745, 48746, 48801, 48803, 48805, 48806, 48807, 48810, 48811, 48812, 48813, 48814, 48824, 48826, 48828, 48829, 48830, 48832, 48836, 48838, 48839, 48863, 48864, 48865, 48867, 48883, 48884, 48885, 48888, 48891, 48892, 48893, 48894, 48895, 48897, 48899, 48900, 48901, 48902, 48903, 48935, 48938, 48940, 48943, 48946, 48947, 48949, 48951, 48952, 48971, 48976, 48982, 48983, 48984, 48985, 48986, 48987, 48988, 48989, 48990, 48991, 49019, 49026, 49027, 49029, 49030, 49031, 49057, 49059, 49060, 49061, 49062, 49063, 49064, 49065, 49066, 49067, 49068, 49069, 49070, 49071, 49072, 49073, 49098, 49099, 49100, 49110, 49111, 49113, 49114, 49116, 49117, 49118, 49121, 49125, 49126, 49127, 49128, 49132, 49133, 49134, 49135, 49136, 49137, 49138, 49139, 49140, 49141, 49142, 49143, 49144, 49155, 49157, 49158, 49159, 49165, 49168, 49217, 49218, 49219, 49224, 49226, 49227, 49228, 49229, 49230, 49232, 49233,

49240, 49241, 49244, 49245, 49246, 49247, 49248, 49249, 49250, 49251, 49253, 49254, 49262, 49263, 49264, 49265, 49266, 49267, 49268, 49270, 49307, 49308, 49309, 49310, 49311, 49312, 49313, 49314, 49316, 49317, 49324, 49329, 49330, 49331, 49332, 49337, 49339, 49340, 49341, 49342, 49345, 49346, 49347, 49348, 49349, 49351, 49352, 49353, 49356, 49358, 49359, 49360, 49361, 49362, 49365, 49366, 49367, 49368, 49369, 49373, 49374, 49376, 49377, 49378, 49379, 49380, 49381, 49382, 49383, 49384, 49385, 49389, 49390, 49432, 49433, 49434, 49435, 49437, 49438, 49439, 49440, 49445, 49446, 49447, 49448, 49449, 49451, 49452, 49453, 49454, 49455, 49456, 49457, 49464, 49466, 49468, 49652, 49653, 49655, 49656, 49657, 49658, 49659, 49660, 49661, 49662, 49664, 49665, 49666, 49736, 49737, 49742, 49743, 49749, 49751, 49752, 49756, 49758, 49760, 49761, 49762, 49763, 49910, 49914, 49919, 49921, 49922, 49923, 49925, 49926, 49928, 49929, 49930, 49932, 49934, 49959, 49960, 49961, 49962, 49964, 49965, 49966, 49967, 49968, 49969, 49970, 50033, 50034, 50036, 50045, 50046, 50047, 50048, 50067, 50068, 50070, 50073, 50074, 50075, 50076, 50077, 50343, 50376, 50377, 50380, 50463, 50464, 50465, 50466, 50467, 50468, 50470, 50471, 50483, 50491, 50492, 50493, 50496, 50497, 50500, 50501, 50517, 50520, 50522, 50526, 50527, 50528, 50529, 50530, 50531, 50535, 50536, 50545, 50546, 50548, 50549, 50550, 50551, 50552, 50553, 50554, 50558, 50560, 50561, 50562, 50563, 50564, 50565, 50569, 50570, 50596, 50597, 50598, 50599, 50600, 50601, 50602, 50603, 50604, 50605, 50606, 50607, 50613, 50614, 50615, 50616, 50617, 50650, 50654, 50655, 50656, 50658, 50660, 50661, 50662, 50663, 50664, 50665, 50666, 50667, 50668, 50669, 50670, 50671, 50672, 50673, 50674, 50675, 50676, 50677, 50678, 50679, 50853, 50854, 50857, 50862, 50863, 50874, 50875, 50876, 50885, 50886, 50887, 50889, 50897, 51015, 51093, 51152, 51154, 51159, 51160, 51162, 51163, 51165, 51166, 51167, 51171, 51172, 51175, 51176, 51177, 51178, 51179, 51180, 51181, 51182, 51183, 51187, 51188, 51200, 51201, 51202, 51206, 51208, 51209, 51210, 51211, 51212, 51213, 51214, 51215, 51217, 51220, 51222, 51225, 51226, 51228, 51229, 51230, 51234, 51236, 51237

Appendix B: External Photon Conversion in sPHENIX Detector

B.1 Overview of External Conversion

In heavy ion collision, precision of EMCal measurements relies on the accurate reconstruction of photons originating from the primary collision vertex. However, before reaching the EMCal, these photons must traverse the internal detector subsystems like **M**onolithic Active Pixel Sensors **V**er**T**e **X**Detector (MVTX), **INT**ermediate Silicon **T**racker (INTT), **T**ime **P**rojection **C**hamber (TPC) and the beryllium beam pipe. These subsystems and their mechanical support structures represent a significant material budget. Any photon traversing this material possess a non-negligible probability of undergoing pair production ($\gamma \rightarrow e^+ + e^-$) before interacting with the EMCal. In addition, the sPHENIX detector operates within a 1.4 T (Tesla) magnetic field which deflects those converted e^+e^- pairs significantly. Such deflection, sometimes, can create two distinct and well separated EMCal clusters. Since, external conversion affects the clusters, they should be corrected accounted for our study of neutral mesons.

B.2 Neutral Meson Reconstruction

As we analyzed neutral meson via two photon decay mode, we expect two EMCal clusters: one for each photon. As external conversion of photon creates more than two clusters, thus, introduces a systematic effect in our analysis. The most robust way of analyzing external conversion utilized the track related information. Since such information were not available for our data, we estimated the external conversion via the number of EMCal clusters in simulation. For this, we simulated single meson decay per event and pass through the full **GEANT4** description of the sPHENIX detector. In reconstructed data, if there exists more than two clusters, those meson certainly have undergone external conversion. If only one

cluster is present, we ignore that meson in our analysis, as they could be associated with efficiency and not external conversion.

The situation becomes nontrivial when exactly two clusters were present. To study this subtle situation, we developed a conservative method to identify if either of the photon from parent meson have converted after interacting with detector. First, we matched each cluster to one of the two truth photons by choosing the closest in ΔR. Such matching must be unique in a sense there must be one photon to one cluster matching. Additional criteria was that every matches must be sufficiently close in angular distance ($\Delta R \leq 0.5$) and energy asymmetry between photon and cluster must be $E_\alpha \leq 0.6$. When any of these criteria was not satisfied, the parent meson was considered as suffering from external conversion.

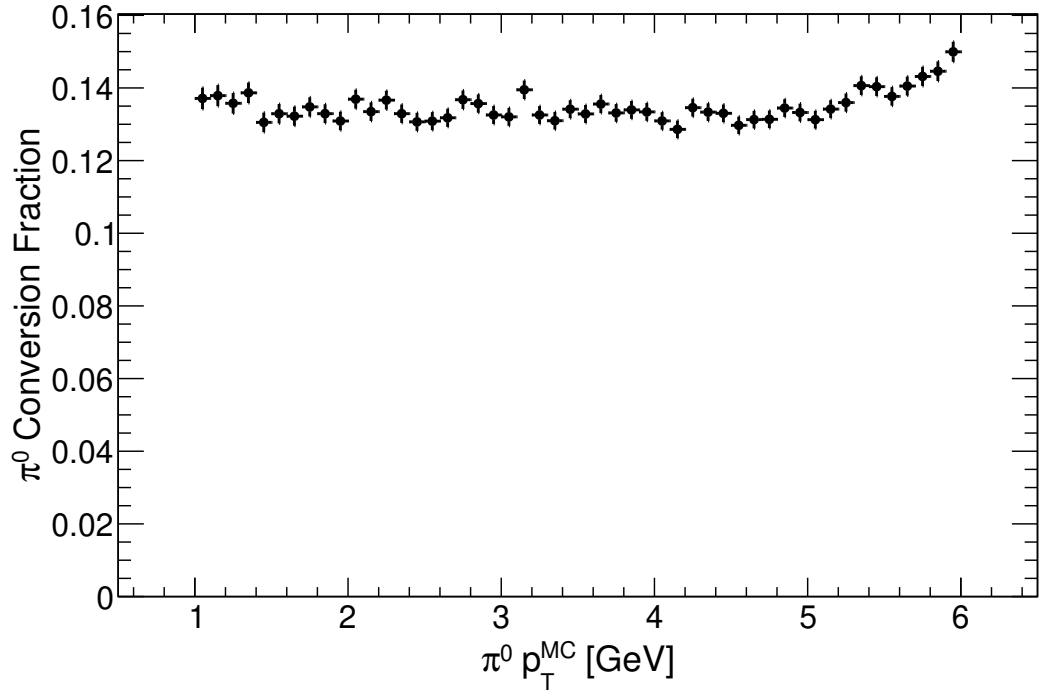


Figure B.1: Study of external photon conversions as a function of transverse momentum (p_T) for $\pi^0 \to \gamma\gamma$, where at least one of the photon gets converted before being detected in EMCal.

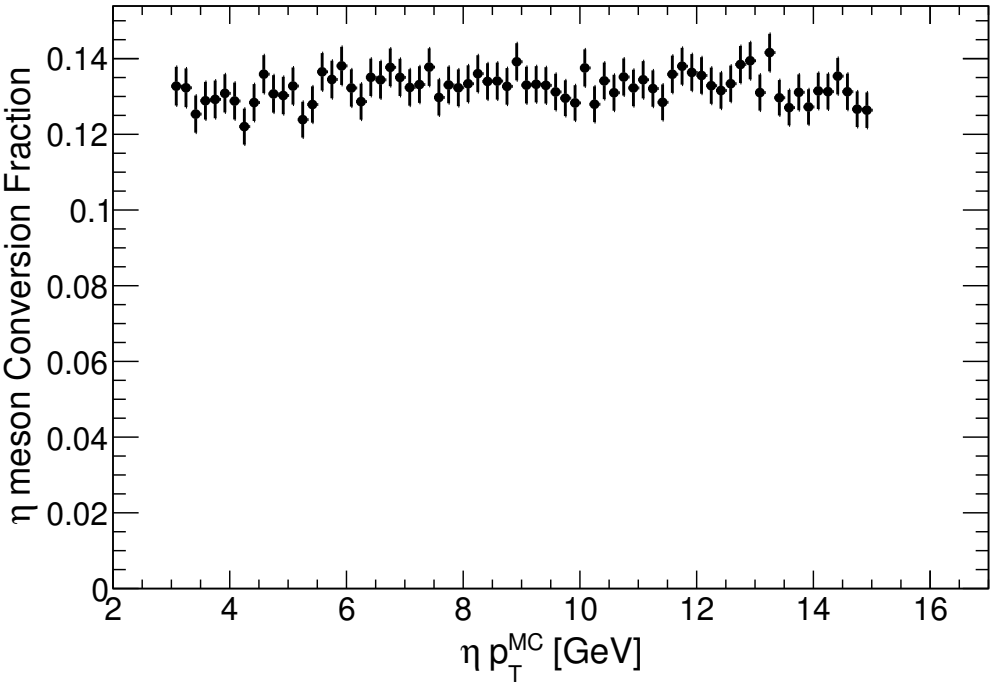


Figure B.2: Study of external photon conversions as a function of transverse momentum (p_T) for η meson decaying to two photons. Here, the conversion fraction reflects the probability that at least one of the decayed photon undergoes conversion.

The figure B.1 and figure B.2 suggest approximately 13 - 14 % of both π^0 and η meson was affected by external conversion which is significant for our study. As this study was a simple conservative estimate, this percentage could be slightly different on using more robust analysis. This conclusion give us a motivation to use side-band fitting method as primary standard method for background estimation.

The external photon conversion was fully modeled within the detector simulation framework, and its impact was systematically propagated while calculating detector efficiency. Therefore, no independent or supplementary study was required to account for this effect for the scope of this study.

Appendix C: Tables of Invariant Cross Section Measurements

Table C.1: The invariant cross section and its associated statistical and systematic uncertainties of π^0 as a function of transverse momentum (p_T), for pseudorapidity range of $-0.35 < \eta < 0.35$.

p_T [GeV]	Invariant Cross Section	Stat. Error	Syst. Error (+)	Syst. Error (−)
1.713	8.2299×10^{-2}	0.1812×10^{-2}	6.6437×10^{-2}	3.6964×10^{-2}
2.218	8.0510×10^{-3}	0.1078×10^{-3}	1.9647×10^{-3}	1.9689×10^{-3}
2.722	1.8119×10^{-3}	0.0214×10^{-3}	0.4695×10^{-3}	0.5406×10^{-3}
3.224	4.8312×10^{-4}	0.0572×10^{-4}	1.2116×10^{-4}	1.4783×10^{-4}
3.727	1.4973×10^{-4}	0.0192×10^{-4}	0.4117×10^{-4}	0.4597×10^{-4}
4.229	5.3093×10^{-5}	0.0830×10^{-5}	1.3747×10^{-5}	1.6158×10^{-5}
4.730	2.2537×10^{-5}	0.0453×10^{-5}	0.5493×10^{-5}	0.6754×10^{-5}
5.232	9.2706×10^{-6}	0.2674×10^{-6}	2.2252×10^{-6}	2.7923×10^{-6}
5.733	4.1143×10^{-6}	0.0424×10^{-6}	0.9843×10^{-6}	1.2369×10^{-6}
6.235	2.1981×10^{-6}	0.0242×10^{-6}	0.5299×10^{-6}	0.6587×10^{-6}
6.736	9.9656×10^{-7}	0.1251×10^{-7}	3.3433×10^{-7}	3.2794×10^{-7}
7.237	5.2737×10^{-7}	0.0789×10^{-7}	1.6349×10^{-7}	1.7722×10^{-7}
7.738	3.4761×10^{-7}	0.0598×10^{-7}	0.8770×10^{-7}	1.6335×10^{-7}